| INCH-POUND |

MIL-D-80002E
9 August 1989
SUPERSEDING
MIL-D-80002D
21 October 1982

MILITARY SPECIFICATION

DRILLING MACHINE, UPRIGHT FLOOR TYPE, ROUND AND BOX COLUMN, SLIDING HEAD, PLAIN TABLE, POWER FEED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers electric motor driven, floor mounted, round and box column, sliding head, power feed, single and multiple spindle, upright drilling machines with equipment as described herein.

1.2 Classification. This specification covers drilling machines of the following types, and sizes. The type style, and size required for specific procurement shall be as specified (see 6.2). All sizes listed below are available in type I and type II machines.

Type I - Round or Box Upper Column, Round Lower Column
  Style A - One spindle

Type II - Round or Box Upper Column, Box Lower Column
  Style A - One spindle
  Style B - Two spindles
  Style C - Three spindles
  Style D - Four spindles
  Style E - Five spindles
  Style F - Six spindles

| Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSM, 2163 Airways Boulevard, Memphis, Tennessee 38114-5051, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter. |

AMSC N/A FSC 3413
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:31Z
Check the source to verify that this is the current version before use.

Size 22 - 1-1/4 inch drill capacity - 22 inch swing
Size 24 - 2 inch drill capacity - 24 inch swing
Size 26 - 2-1/2 inch drill capacity - 26 inch swing
Size 28 - 3 inch drill capacity - 28 inch swing

2. APPLICABLE DOCUMENTS.

2.1. Government documents.

2.1.1 Specification, standards and handbook. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of.

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

FED-STD-376 - Preferred Metric Units for General Use by the Federal Government.

MILITARY

MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interferences.

MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.

MIL-STD-961 - Military Specifications and Associated Document, Preparation of.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publication and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-0001

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:31Z
Check the source to verify that this is the current version before use.

CODE OF FEDERAL REGULATIONS (CFR)

U. S. DEPARTMENT OF LABOR, OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)

29 CFR 1910 - Occupational Safety and Health Standards.

(Application of copies should be addressed the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402-0001.)

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific directed by the contracting activity.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issue of the documents which are DOD adopted are those listed in the issue of the DODISS cite in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN GEAR MANUFACTURERS' ASSOCIATION (AGMA)

AGMA 360.02 - Manual for Machine Tool Gearing.

AGMA 390.03 - Gear classification, Materials and measuring Methods for Unassembled Gears.

(Application for copies should be addressed to the American Gear Manufacturers' Association, Standards Department, 1500 King Street, Suit 201, Alexandria, VA. 22314-2717.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B5.1 - T-Slots, Their Bolts, Nuts, Tongues, and Cutters. (DOD adopted)

ANSI B11.8 - Drilling, Milling, and Boring Machines, Safety Requirements for the Construction, Care, and use. (DOD adopted)

ANSI/ISO 54 - Cylindrical Gears For General Engineering and for Heavy Engineering, Modules and Diametral Pitches of.

(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York, NY 10018-3363.)

NATIONAL ELECTRICAL MANUFACTURERS' ASSOCIATION (NEMA)

NEMA MG 1 - Motors and Generators. (DOD adopted)

NEMA ICS 1 - Industrial Control and Systems. (DOD adopted)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:31Z
Check the source to verify that this is the current version before use.

(Application for copies should be addressed to the National Electrical Manufacturers' Association, 2101 L Street, N.W., Washington, DC 20037-1580.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 79 - Electrical Standard for Industrial Machinery. (DOD adopted)

(Application for copies should be addressed to the National Fluid Power Association, Batterymarch Park, Quincy, MA 02269-9101.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.4.

3.2 Design. The Drilling machine shall be new (not a prototype) and one of the manufacturer's current production models capable of operations in accordance with the requirements herein. The drill shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be readily accessible for adjustment, replacement, and repair.

3.2.1 Measuring and indicating device calibrations. Measuring and indicating devices shall be graduated in inch-pounds (see 6.4.1), metric (see 6.4.2), or dual (inch-pound and metric) units as specified by the procuring activity (see 6.2). Regardless of the measurement system used, all measuring and indicating devices on the drill shall be graduated in the same system.

3.2.1.1 Dials. All dials used to indicate machine axes position or tool movement shall be graduated in increments not larger than 0.001 inch. Dial land handwheel circumferences shall be permanently and legibly engraved or etched on a nonglare background with graduations that can be read from the operator's normal position. All feed dials shall be calibrated in such a manner that the last dial graduation progresses into and is continuous with the first dial graduation as the dial is rotated through the zero position.

3.2.2 Reclaimed materials. The machine may contain reclaimed materials to the maximum extent possible provided such materials will not jeopardize the intended use, performance, or design life of the machine. Reclaimed materials shall have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:31Z
Check the source to verify that this is the current version before use.

the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification.

3.2.3 Energy efficiency. The drill and its components that directly consume energy in normal operation shall be designed and constructed for energy efficiency as governed by the developments available within the industry.

3.2.4 Controls. All mechanical, hydraulic, and pneumatic operating controls shall be located convenient to the operator's work station(s).

3.2.5 Safety and health requirements. All parts, components, mechanisms, and assemblies furnished on the drill, whether or not specifically required herein, shall conform to all requirements of OSHA 29 CFR 1910 and ANSI B11.8. If a conflict arises between 29 CFR 1910 and ANSI B11.8, 29 CFR 1910 shall apply. Additional safety and health requirements shall be as specified (see 6.2). Covers, guards, or other safety devices normally furnished as standard on the manufacturer's commercial drills supplied to the commercial market shall be provided for the point of operation and all other parts of the drill that present safety hazards. Additional guarding dependent on workpiece size and configuration shall be provided when specified and fully described by the procuring activity (see 6.2).

3.2.6 Mercury restriction. The drill shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture.

3.2.7 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the drilling machine.

3.2.8 Environmental protection. During operation, service, transportation, and storage of the machine, it shall not emit materials hazardous to the ecological system as prohibit by Federal, state, or local statutes in effect at the point of installation.

3.2.9 Lubrication. Means shall be provided to ensure adequate lubrication for all moving parts. Recirculating lubricating systems shall include a filter which is cleanable or replaceable and shall have a low lubrication level warning light that is in clear view of the operator's normal work station. An automatic lubrication system shall be provided for all ways and axis screws. Each lubricant reservoir shall have at least a 24 hour capacity and means for determining fluid level. All oil holes, grease fittings, and filer caps shall be readily accessible.

3.2.10 Interchangeability. To provide for replacement of worn parts, all parts shall be manufactured to definite dimensions and tolerances that will permit installation of replacement parts without modification of the part or machine.

3.3 Construction. The drilling machine shall be constructed of parts which are new, without defects, and free of repairs. The structure shall withstand all forces encountered during operation of the drilling machine to its maximum rating and capacity without permanent distortion.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:31Z
Check the source to verify that this is the current version before use.

3.3.1 Castings and forgings. All castings and forgings shall be free of defects, scale, and mismatching. No processes such as welding, peening, plugging, or filling with solder or paste shall be used for reclaiming any defective part. Such processes may be used only for enhancing surface finish and appearance.

3.3.2 Fastening devices. All fasteners shall be installed to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be permanently installed.

3.3.3 Surfaces. All surfaces shall be clean and free of harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.4 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these operations shall be employed as a repair measure for any defective part.

3.3.5 Painting. Unless otherwise specified (see 6.2), the drilling machine shall be painted in accordance with the manufacturer's commercial practice and color.

3.3.6 Threads. All threaded parts used on the machine and its related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

3.3.7 Gears. All gears shall be machined from a material suitable for the intended purpose. All gears shall be heat treated by a process that will impart the hardness and toughness that will enable the gear train to transmit full rated torque of the drive motor without gear damage, failure,or premature wear.

3.3.8 Electromagnetic interference control. When specified (see 6.2), equipment furnished under this specification shall comply with MIL-STD-461. The equipment and subsystem(s) class and the emission and susceptibility requirements shall be as specified.

3.4 Components. The drilling machine shall consist of not less than the following components.

3.4.1 Base. The base shall be an iron or iron alloy casting, or a steel weldment, ribbed and braced to provide a foundation for the entire machine. Means shall be provided for leveling and securing the machine to a flat mounting surface. The coolant fluid sump or reservoir, as required under 3.4.6, shall be included as an integral part of the base or as a separate external unit. That portion of the base that is below the spindle/quill assembly on the type I machine, shall also serve as an auxiliary working surface for accommodating extra large workpiece sizes. The auxiliary working surface shall be machined to not rougher than 125 microinch finish and have at least two parallel T-slots. Machined T-slots shall be in accordance with ANSI

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:31Z
Check the source to verify that this is the current version before use.

B5.1. The periphery of the auxiliary working surface shall have a coolant fluid trough.

3.4.2 Column. The column shall be mounted on the base and reinforced to provide support and minimize deflection. Machines with cylindrical columns shall be cross sectionally round on the lower column with the exterior surface machined for vertical adjustments of the work table; and the upper columns shall be cross sectionally round with the exterior surface machined for vertical adjustment of the sliding head. Machines with box columns shall have machined guideways for vertical adjustment of the work table and the upper columns shall be sectionally round with the exterior surface machined for vertical adjustment of the sliding head or rectangular with machined guide ways for vertical adjustment of the sliding head.

3.4.3 Top head and spindle drive. Unless otherwise specified (see 6.2), the top head shall house the spindle drive assembly in a stationary position at the top of the column. The spindle shall be driven by an assembly of motor-driven mechanical components with shafts, gears, belts, or pulleys as necessary for developing and maintaining spindle rotating speeds and feed rates as required herein. All rotating and sliding components shall be mounted on bearings or bushings with means for positive and continuous lubrication.

3.4.4 Sliding head. The sliding head shall be an iron or iron alloy casting or a steel weldment, arranged to support the spindle/quill assembly. The head shall be mounted to the column and be vertically adjustable. Means shall be provided for adjustment to take up wear. A locking device shall be provided to lock the sliding head in position after adjustment. The sliding head shall be counterbalanced or vertically adjusted be the use of a high mechanical advantage drive to facilitate ease of adjustment.

3.4.5 Spindle/quill assembly. The quill shall be machined of steel allow, and shall be fitted to the sliding head. Vertical travel of the quill shall be accomplished by means of a rack and pinion drive. Quill position shall be indicated by either a graduated scale engraved on the quill in increments of not more than 1/8 inch, or by a feed depth dial. The spindle shall be machined from steel alloy and shall rotate within the quill on boll or roller bearings. The characteristics of the spindle shall comply with ANSI B5.10. The quill/spindle assembly shall be counterbalanced and be power fed for accomplishing performance specified herein. The spindle feed shall be obtained through a variable speed motor drive or a gear drive arrangement which is hand lever operated 5through a positive engagement type clutch.

3.4.6 Worktable. The worktable shall be an iron or iron allow casting, bibbed and reinforced to withstand maximum capacity workpiece weights and tool thrust loads. The working surface of the worktable shall be machined to a finish of not rougher than 125 microinches, level, and of the size required herein. Worktable height adjustment shall be accomplished by a mechanical drove arrangement of either a crank and worm gear, crank and screw-jack or crank and rack and opinion type mechanism. The working surface of the worktable for the type I machine shall be round as viewed from the top, rotatable around its center, and revolvable around the column to a position

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:31Z
Check the source to verify that this is the current version before use.

necessary to clear the auxiliary working surface of the base. The working surface of the worktable for the type II machine shall be rectangular with a coolant fluid trough around its periphery with means for returning coolant to the reservoir. Means shall be provided for securing the worktable in any position within its adjustment, rotating or revolving ranges. The working surface of the worktable shall have the number and arrangement of T-slots normally furnished by the manufacturer. Machined T-slots shall be in accordance with ANSI-B5.1

3.4.7 Coolant system. Unless otherwise specified (see 6.2), a coolant system shall be provided. The coolant system shall include a sump or reservoir, a power driven pump, and piping. The coolant reservoir shall be of sufficient capacity to permit full flow cooling. The cooling system shall include a baffle(s) or strainer which is easily removable for cleaning. The reservoir shall be equipped with means of emptying and cleaning. A coolant delivery nozzle and valve shall be provided to permit the operator to direct and control the amount of coolant flow. Means for returning coolant from work area to the reservoir shall be provided.

3.4.8 Electrical system. The electrical system shall be wired to conform to NFPA 79. Unless otherwise specified (see 6.2), each machine shall wired to draw all of its electrical power from a single 230/460-volt, 3-phase, 60-hertz (Hz) circuit. Unless otherwise specified (see 6.2), the drilling machine shall be initially wired for operation on 460 volts.

3.4.8.1 Motors. Motors shall meet the requirements of 3.4.8. Motors shall be rated for 60 minute duty cycle and shall have ball or roller bearings of the sealed and permanently lubricated type. Unless otherwise specific (see 6.2), alternating-current (ac) motors shall be designed to operate on 60-Hz. The motor enclosure shall meet the requirements for a dripproof enclosure. All motors shall conform to the requirements of NEMA MG-1, and table I.

3.4.8.2 Worklight. Unless otherwise specified (see 6.2), a worklight shall be provided. The light shall have a protective shield, an on-off switch, and shall be adjustable to reflect maximum light on maximum light on the work area. If a built-in lighting system is provided, the adjustable light described above shall not be required.

3.4.8.3 Hour meter. When specified (see 6.2), the drilling machine shall be equipped with an hour meter. The hour meter shall be installed to display accumulated operating time of the main drove motor. The meter shall be of the nonresitting type and shall have a range of 0 to 9,999 hours in increments of not grater than 1 hour. The meter shall be sealed to prevent the entrance of dust and moisture and shall be mounted to withstand shock and vibration generated by the machine.

3.5 Size and capacity. Unless otherwise specified (see 6.2), the size and capacity of the machines shall be not less than specified in table I for the machine size specified.

3.6 Alignment tolerances. Each drilling machine shall meet the alignment tolerances specified in table II.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:31Z
Check the source to verify that this is the current version before use.

3.7 Performance. The machine shall be capable of drilling holes through metal solids to include cast iron for accomplishing maximum capacity hole sizes, and spindle speed and feed rate required by table I.

3.8 Standard equipment. All standard equipment normally provided with the manufacturer's commercial machine shall be furnished.

3.9 Optional equipment. Optional equipment shall be furnished as specified and fully described (see 6.2).

3.10 Repair parts. Required repair parts shall be furnished as specified (see 6.2).

3.11 Marking on instruments, control panels, charts, and plates. All words on instruments, control panels, charts, and plates shall be in the English language. Characters shall be permanently marked in boldface on a contrasting background. All plates shall be corrosion resistant.

3.11.1 Lubrication plate. When specified (see 6.2), a lubrication plate or chart shall be attached to each machine. Unless specified (see 6.2), the information provided on the chart or plate shall be as listed below.

Points of lubrication application
Servicing interval
Type of lubricant
Federal or military specification for each lubricant (if applicable)

3.11.2 Nameplate. A nameplate shall be securely attached to each machine. Unless otherwise specified (see 6.2), the nameplate shall provide the information listed below.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Amp rating of largest motor
Short circuit/over-current protection rating
Contract number or order number
National Stock Number or Plant Equipment Code
Date of manufacture

3.11.3 Workmanship. Workmanship of the drilling machine and its accessories shall meet all requirements specified herein and shall be of a quality equal to that prevailing among manufacturers producing equipment of the type covered by this specification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:31Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations am tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractors overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submissions of known defective material. either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows.

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed in accordance with the test conditions specified in 4.7.1.

4.4 First article inspection. When a first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2), first article inspection shall pass the first article examination and all test to be accepted.

4.5 Quality conformance inspection. Quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2), quality conformance inspection shall consist of the examination in 4.6, the tests in 4.7, and the inspection in 4.3. The machine shall pass the examination, all test, and the inspection to be accepted.

4.6 Examination. The drilling machine shall be examined to determine compliance with all requirements of this specification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:31Z
Check the source to verify that this is the current version before use.

4.7 Tests.

4.7.1 Test conditions. All tests shall be performed in an indoor facility with ambient conditions of 41 degrees to 104 degrees Fahrenheit and 20 percent to 95 percent relative humidity.

4.7.2 Operational test. The drilling machine shall be operated at no load for not less than 15 minutes. Proper operation of all controls, motors, adjusting mechanisms, and accessories shall be verified during the trial period.

4.7.3 Alignment accuracy tests. Each machine shall be tested to determine that it complies with the alignment accuracy requirements of table II.

4.7.4 Performance test. Each machine shall be tested by drilling three holes through a solid, one-inch thick piece of cast iron as follows:

Spindle Speed (not less than)	Spindle Feed (not less than)	Machine drill size	
Size 22	1-1/4 inch	270 RPM	0.009 In/Rev
Size 24	2 inch	150 RPM	0.015 In/Rev
Size 26	2-1/2 inch	120 RPM	0.015 In/Rev
Size 28	3 inch	100 RPM	0.015 In/Rev

Failure of the machine to maintain the above specified minimum spindle speed and feed rate during the performance test, or any permanent distortion of machine components resulting from the test, shall be cause for rejection.

4.7.5 Electromagnetic interference control test. Equipment requiring electromagnetic interference control shall be tested for compliance with 3.3.8 using the procedures given in MIL-STD-462.

4.8 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Packaging Requirements. The requirements for packaging shall be in accordance with MIL-M-18058.

6. NOTES

6.1 Intended use. The drilling machines covered by this specification are intended for use in production shops where wide varieties of drilling operations are performed.

6.2 Ordering data.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:31Z
Check the source to verify that this is the current version before use.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type, class, and size required (see 1.2).

c. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual document referenced (see 2.1.1 and 2.2).

d. First article, if required (see 3.1).

e. System of units for measuring and indicating devices (see 3.2.1).

f. Additional safety and health requirements, if required (see 3.2.1).

g. Specify configuration of additional point-of-operation guarding, if required (see 3.2.5).

h. Painting, it different (see 3.3.5).

i. If electromagnetic interference control is required, specify the equipment and subsystem class and the emission and susceptibility required (see 3.3.8).

j. Type spindle drive, if different (see 3.4.3).

k. Cooling system if different (see 3.4.7)

l. Electrical system, if different(see 3.4.8.

m. Initial voltage, if different (see 3.4.8).

n. Motors, if different (see 3.4.8.1).

o. Worklight, if not required (see 3.4.8.2).

p. Hour meter, if required (see 3.4.8.3).

q. Size and capacity, if different (see 3.5).

r. Specify and fully describe optional equipment, if required (see 3.9).

s. Specify applicable repair parts, if required (see 3.10).

t. Lubrication plate, if required (see 3.11.1).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:31Z
Check the source to verify that this is the current version before use.

u. Information to be provided on lubrication plate if different (see 3.11.1).

v. Nameplate, if different (see 3.11.2).

w. Inspection condition, if different (see 4.3).

x. First article inspection, if different (see 4.4).

y. Quality conformance inspection, if different (see 4.5).

6.2.2 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Description (DIDs) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DIDs are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	Did Title	Suggested Tailoring
.	.	.	.
.	.	.	.

The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DIDs are cited on the DD Formal 1423.

6.2.2.1 Technical data. All technical data ordered using a DD Form 1423 should be written in the English language.

6.3 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a production item, a sample selected from the first____ production items, a standard production item from the contractors current inventory (see 3. ___), and the number of items to be tested as specified in 4.__. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Definitions.

6.4.1 Inch-pound units. Inch-pound units are a system of measures based on the yard and pound commonly used in the United States of America and defined

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:31Z
Check the source to verify that this is the current version before use.

by the National Bureau of Standards. Inch-pound units having the same names in other countries may differ in magnitude.

6.4.2 Metric units. Metric units are a system of basic measures defined by the International System of Units based on "Le System International D'Units (SI)," of the International Bureau of Weights and Measures. These units are described in ASTM E 380 and IEEE 268 (MIL-STD-376).

6.5 Measurement system. In this specification, all measurements, dimensions,sizes, and capacities are given in inch-pound units. These measurements may be converted to metric units through the use of the conversion factors and methods specified in FED-STD-376.

6.6 Changes from previous issues. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.7 Safety and health requirements. 29 CFR 1910 limits only the total hazard level (noise, radiation, electromagnetic emissions, noxious vapors, air contaminants, and heat) of the environment in which a machine will operate. It does not limit the hazard level of individual machines in an operating environment. The procuring activity should analyze the existing hazard level in the proposed operating environment and specify additional requirements necessary to integrate this new machine into its future environment.

6.8 Training. Training required by the procuring activity should be provided as specified in the contract.

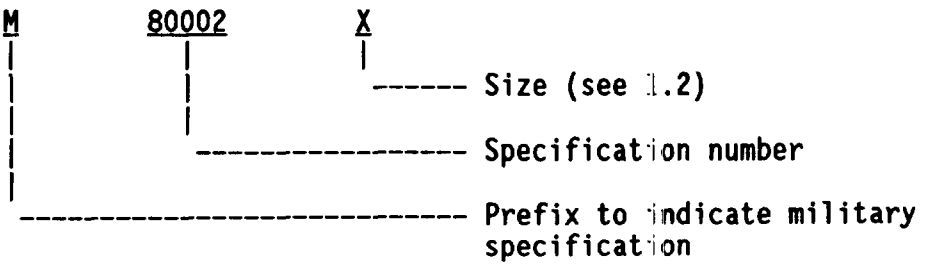
6.9 Warranty. Warranty requirements should be as specified by the procuring activity in the contract.

6.10 Inspection location. The contractor should identify in his response to the solicitation the location where inspection and tests are to be performed.

6.11 Subject term (keyword) listing.

Drilling machine
Box column
Power feed
Plain table
Round and box column
Sliding head

6.12 Part or Identifying Number (PIN). The PIN to be used for machines acquired to this specification are created as follows:



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:31Z
Check the source to verify that this is the current version before use.

TABLE I. Sizes and capacities type I

Characteristic	Requirements			
	22	24	26	28
Drilling capacity, cast iron, inches	1-1/4	2	2-1/2	3
Distance of face of round column to spindle center, inches	11	12	13	14
Swing drills to center of circle, diameter, inches	22	24	26	28
Distance from spindle nose to table, inches	23	23	24	25
Distance from spindle nose to finished surface of base, inches	41	41	41	41
Vertical travel of spindle/quill assembly inches	5	5	8	8
Working surface of base, width x length inches	16x16	20X22	20x22	20x22
Number of feeds	4	4	4	4
Rates, spindle feed, inches per revolution	to .016	to .018	to .018	to .016
Number of spindle speeds	4	4	4	4
Spindle speed range, rpm	65 to 1350	110 to 1100	80 to 1000	80 to 1000
Diameter, round column, inches	5-1/2	6-1/2	8	8
Spindle nose, diameter, inches	1-5/8	2	2-1/2	2-1/2
Spindle bored to fit morse taper	4	4	5	5
Diameter quill, inches	2-1/2	2-9/16	2-3/4	3-1/2
Table diameter, inches	17	23	23	23

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:31Z
Check the source to verify that this is the current version before use.

TABLE I. Cont. Size and capacities type II

Characteristic	22	24	26	28
Drilling capacity, cast iron, inches	1-1/4	2	2-1/2	3
Distance box-column ways to spindle center, inches	11	12	13	13
Swing, drills to center of circle, diameter, inches	22	24	26	28
Distance from spindle nose to table, inches	23	23	24	25
Vertical travel of spindle/quill assembly, inches	5	5	8	8
Number of feeds	4	4	4	4
Rates, spindle feed, inches per revolution	.004 to .016	.006 to .018	.006 to .018	.006 to .018
Number of spindle speeds	4	4	4	4
Spindle speed range, rpm	65 to 1350	100 to 1100	80 to 1000	80 to 1000
Spindle nose, diameter, inches	1-5/8	2	2-1/2	2-1/2
Spindle bored to fit morse taper	No.4	No.4	No.5	No.5
Diameter quill, inches	2-1/2	2-9/16	2-3/4	2-3/4
Table working surface, width x length, inches				
Style A	22x22	19x22	20x22	23x22
Style B	22x38	19x36	20x36	23x46
Style C	22x54	19x56	20x58	23x58
Style D	22x70	19x72	20x76	23x77
Style E	22x86	18x88	20x90	23x90
Style F	22x102	19x104	20x110	23x115
Distance between spindle centers inches	19	19	19	19
Motor, horsepower	3	5	7-1/2	7-1/2

NOTE: Requirements are not less than those shown. When a range is shown, the required performance is from the stated minimum or less to the stated maximum or greater.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:31Z
Check the source to verify that this is the current version before use.

TABLE II. Alignment and accuracy tolerances

Characteristic	Maximum Allowable Deviation
Working surface of base, flat (type I style A)	.0016"/12"
Working surface of worktable, flat	.0016"/12"
Column guideways square with working surface of the worktable, in longitudinal and transverse planes	.003"/12"
Spindle taper, runout 1 inch from nose	.001" TIR
Spindle taper, runout 6 inches from nose	.002" TIR
Spindle square with working surface of the worktable, turn around method with 6-inch radius, in longitudinal and transverse planes	.003"/12"

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:31Z
Check the source to verify that this is the current version before use.

Custodians:
Army - AL
Navy - SH
Air Force - 99

Review activities
Navy - SH
Air Force - 84

User activities:
Army - SM
Navy - YD, MC

Preparing activity
DLA - IP

Project (3413-0144)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:31Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-D-80002E	Drilling Machine, Upright Floor Type, Round and Box Column, Sliding Head, Plain Table, Power Feed

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: –

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

Source: https://assist.dla.mil – Downloaded: 2026-10-23T04:21Z
Check the source to verify that this is the current version before use.